Chapter Contents

Chapter 8 - Linear Solution Theory

This chapter studies what can be known about solutions of linear ordinary differential equations before a specific solution method is chosen.

The emphasis is structural. We will ask why homogeneous solutions can be combined, how many independent solutions are needed, what the Wronskian actually tests, and why one particular forced response generates an entire nonhomogeneous family.

The central questions are:

- what makes an equation linear at order n?
- what do existence and uniqueness guarantee?
- why does an nth-order homogeneous equation require n independent solution directions?
- how can linear independence be tested directly or with a Wronskian?
- why is a nonhomogeneous solution always "complementary part plus one particular part"?

The chapter's organising idea is:

linearity makes the solution set predictable: homogeneous solutions form a space of reusable directions, and forcing shifts that space to a new family.

Recognising A Linear Equation Of Any Order

The Goal Of This Block

The goal is to recognise an nth-order linear differential equation and separate four ideas that are often confused:

- order
- linearity
- homogeneity
- constant or variable coefficients

these labels answer different questions, so an equation may carry several of them at once.

The General Linear Pattern

An nth-order linear ordinary differential equation has the form:

$$\boxed{b_n(x)y^{(n)} + b_{n-1}(x)y^{(n-1)} + \cdots + b_1(x)y' + b_0(x)y = g(x)}$$

where:

- $y = y(x)$ is the unknown function
- $y^{(k)}$ is its kth derivative
- each coefficient $b_k(x)$ depends only on x
- the forcing term $g(x)$ also depends only on x
- the leading coefficient $b_n(x)$ is not identically zero

Linearity concerns how the unknown function and its derivatives appear.

They may:

- be added
- be multiplied by functions of x
- appear to the first power

They may not:

- multiply one another
- appear inside nonlinear functions

- the unknown and its derivatives may not be raised to nonlinear powers
- occur in a coefficient that depends on y

The practical meaning is:

the coefficient functions may be complicated in x; the equation is linear only if its dependence on $y, y', \dots, y^{(n)}$ is linear.

Order And Linearity Are Separate

The **order** is the highest derivative appearing in the equation.

Consider:

$$(1 + x^2)y''' - e^x y' + 4y = \sin x.$$

The highest derivative is y''', so the equation is third-order.

It is also linear because:

- y''', y', and y appear only to the first power
- their coefficients $1 + x^2$, $-e^x$, and 4 depend only on x
- the forcing $\sin x$ does not depend on y

By contrast:

$$y'' + yy' = 0$$

is second-order but nonlinear because y multiplies y'.

A second nonlinear example is:

$$y'' + \sin(y) = x$$

is also second-order and nonlinear because the unknown appears inside $\sin(y)$.

Homogeneous And Nonhomogeneous

A linear equation is **homogeneous** when its forcing term is zero:

$$\boxed{b_n y^{(n)} + \cdots + b_1 y' + b_0 y = 0}.$$

It is **nonhomogeneous** when the forcing term is not identically zero:

$$\boxed{b_n y^{(n)} + \cdots + b_1 y' + b_0 y = g(x), \qquad g \not\equiv 0}.$$

For example:

$$x^2 y'' - 3xy' + 2y = 0$$

is homogeneous, whereas:

$$x^2 y'' - 3xy' + 2y = \ln x$$

is nonhomogeneous on $x > 0$.

Here **homogeneous** does not mean the first-order substitution pattern from Chapter 4. In this chapter, it means that a linear operator sends y to zero.

Constant And Variable Coefficients

The constant-coefficient equation:

$$2y''' - 5y' + 7y = e^x$$

has constant coefficients because 2, 0, -5, and 7 are constants.

The variable-coefficient equation:

$$(1 + x^2)y'' + xy' - 4y = 0$$

has variable coefficients because $1 + x^2$ and x vary with the independent variable.

Both equations are linear. Constant coefficients describe a special class of linear equations; they do not define linearity itself.

Normal Form And The Working Interval

If $b_n(x) \neq 0$ on an interval I, divide every term by $b_n(x)$:

$$y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = f(x),$$

where:

$$a_k(x) = \frac{b_k(x)}{b_n(x)}, \qquad f(x) = \frac{g(x)}{b_n(x)}.$$

This is the **normal form**.

The division matters because it may split the real line into separate working intervals.

For example, start from:

$$(x - 2)y'' + (x + 1)y' - y = e^x.$$

The leading coefficient is $x - 2$. It vanishes at $x = 2$.

For $x \neq 2$, divide every term by $x - 2$:

$$\boxed{y'' + \frac{x+1}{x-2}y' - \frac{1}{x-2}y = \frac{e^x}{x-2}}.$$

The natural intervals are:

$$(-\infty, 2) \qquad \text{and} \qquad (2, \infty).$$

The equation cannot be treated as one regular normal-form problem across $x = 2$.

A Classification Workflow

Given an equation, ask the following questions in order.

Step 1: What is the highest derivative?. This determines the order.

Step 2: How does the unknown appear?. Check $y, y', \ldots, y^{(n)}$ for products, powers, or nonlinear functions.

Step 3: Is the forcing zero?. This distinguishes homogeneous from nonhomogeneous linear equations.

Step 4: Do the coefficients vary with x?. This distinguishes constant from variable coefficients.

Step 5: Where is the leading coefficient nonzero?. This determines the intervals on which normal form and the standard theory apply.

Block 1 Summary

An nth-order equation is linear when it can be written as:

$$b_n(x)y^{(n)} + \cdots + b_0(x)y = g(x).$$

Classify order, linearity, homogeneity, and coefficient type separately. Before using the theory, identify an interval on which the leading coefficient does not vanish.

The Linear Differential Operator

The Goal Of This Block

The goal is to package the left side of a linear differential equation into an operator and use its linearity to explain the structure of the solution set.

the operator notation is useful because it exposes which algebraic steps are valid for every linear equation, regardless of order.

Defining The Operator

On an interval where the equation is in normal form, define:

$$\boxed{L[y] = y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y}.$$

The differential equation becomes:

$$L[y] = f(x).$$

The associated homogeneous equation is:

$$L[y] = 0.$$

The same operator L appears in both equations. Only the right side changes.

What Operator Linearity Means

The operator L is linear if, for any sufficiently differentiable functions u and v and constants α and β:

$$\boxed{L[\alpha u + \beta v] = \alpha L[u] + \beta L[v]}.$$

The coefficients α and β must be constants. If they were functions of x, differentiating their products would create extra terms.

Linearity follows from two facts:

$$(\alpha u + \beta v)^{(k)} = \alpha u^{(k)} + \beta v^{(k)}$$

for constant α, β, and multiplication by each coefficient $a_k(x)$ distributes over addition.

Worked Operator Check

Consider the operator:

$$L[y] = y'' + x^2 y' - 3y.$$

Here y is a placeholder for whichever function is supplied to L. The operator takes that input function, forms its first and second derivatives, and combines the three resulting terms according to the displayed rule.

Choose any twice-differentiable functions u and v, and choose any constants α and β. Build a new input function:

$$w = \alpha u + \beta v.$$

The question is whether applying L once to the combined input w gives the same result as applying L to u and v separately and then combining the outputs. In symbols, we want to verify:

$$L[w] = \alpha L[u] + \beta L[v].$$

Step 1: Apply The Operator Rule To w. Replace the placeholder y in the operator definition by w:

$$L[w] = w'' + x^2 w' - 3w.$$

Step 2: Differentiate The Combined Input. Because $w = \alpha u + \beta v$ and α, β are constants:

$$w' = \alpha u' + \beta v',$$
$$w'' = \alpha u'' + \beta v''.$$

Step 3: Substitute All Three Expressions. Use the formulas for w, w', and w'' in $L[w] = w'' + x^2w' - 3w$:

$$\begin{aligned} L[w] &= \alpha u'' + \beta v'' + x^2(\alpha u' + \beta v') - 3(\alpha u + \beta v) \\ &= \alpha u'' + \beta v'' + \alpha x^2 u' + \beta x^2 v' - 3\alpha u - 3\beta v. \end{aligned}$$

Step 4: Group The u-Terms And v-Terms.

$$L[w] = \alpha(u'' + x^2u' - 3u) + \beta(v'' + x^2v' - 3v).$$

Step 5: Recognise The Two Operator Outputs. Applying the original operator rule separately to u and v gives:

$$L[u] = u'' + x^2u' - 3u, \qquad L[v] = v'' + x^2v' - 3v.$$

Therefore the grouped expressions in the previous line are exactly $L[u]$ and $L[v]$:

$$L[w] = \alpha L[u] + \beta L[v].$$

Finally, because $w = \alpha u + \beta v$:

$$\boxed{L[\alpha u + \beta v] = \alpha L[u] + \beta L[v]}.$$

This proves that the operator is linear. Every symbol in the final identity now refers either to one of the two chosen input functions or to one of their constant multipliers.

Why A Nonlinear Operator Fails

Consider instead:

$$N[y] = y'' + y^2.$$

Apply the nonlinear operator $N[y] = y'' + y^2$ to $u + v$:

$$N[u+v] = (u+v)'' + (u+v)^2.$$

Expand both terms:

$$N[u+v] = u'' + v'' + u^2 + 2uv + v^2.$$

But:

$$N[u] + N[v] = u'' + u^2 + v'' + v^2.$$

The extra term $2uv$ means:

$$N[u+v] \neq N[u] + N[v]$$

in general. Superposition cannot be assumed for nonlinear equations.

Block 2 Summary

Writing a linear equation as $L[y] = f$ makes the key identity visible:

$$L[\alpha u + \beta v] = \alpha L[u] + \beta L[v].$$

This identity will explain superposition, complementary solutions, and the form of every nonhomogeneous solution.

Existence, Uniqueness, And Initial Data

The Goal Of This Block

The goal is to understand what the standard existence-and-uniqueness theorem guarantees for a regular nth-order linear initial-value problem.

The theorem does not provide a formula. It tells us whether the initial data select exactly one solution on a stated interval.

The Initial Data For An nth-Order Equation

For an nth-order equation, the standard initial conditions prescribe the function and its first $n-1$ derivatives at one point x_0:

$$\boxed{\begin{aligned} y(x_0) &= \gamma_0, \\ y'(x_0) &= \gamma_1, \\ &\vdots \\ y^{(n-1)}(x_0) &= \gamma_{n-1}. \end{aligned}}$$

There are n pieces of initial data.

For a second-order equation, these are:

$$y(x_0) = \gamma_0, \qquad y'(x_0) = \gamma_1.$$

For a third-order equation, one also specifies $y''(x_0)$.

The number of initial conditions matches the number of arbitrary constants in a general solution.

The Existence-And-Uniqueness Statement

Consider the normal-form equation:

$$y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_0(x)y = f(x).$$

Suppose:

- $a_0, \ldots, a_{n-1}$ are continuous on an interval I
- f is continuous on I
- the initial point x_0 lies in I

Then, for any prescribed values $\gamma_0, \ldots, \gamma_{n-1}$, the initial-value problem has exactly one solution on I.

The theorem contains two promises:

1. **existence:** at least one solution satisfies the equation and data
2. **uniqueness:** no second, different solution satisfies the same equation and data on that interval

The interval is part of the conclusion. Continuity must hold throughout it.

Worked Interval Example

Consider:

$$(x-2)y'' + (x+1)y' - y = e^x.$$

Suppose the initial conditions are prescribed at $x_0 = 3$.

Step 1: Put the equation in normal form. For $x \neq 2$, divide every term by $x - 2$:

$$y'' + \frac{x+1}{x-2}y' - \frac{1}{x-2}y = \frac{e^x}{x-2}.$$

Step 2: Identify where the coefficients are continuous. The coefficient and forcing functions are:

$$a_1(x) = \frac{x+1}{x-2}, \qquad a_0(x) = -\frac{1}{x-2}, \qquad f(x) = \frac{e^x}{x-2}.$$

All three are continuous on:

$$(-\infty, 2) \qquad \text{and} \qquad (2, \infty).$$

Step 3: Choose the interval containing the initial point. Since $x_0 = 3$, the relevant interval is:

$$(2, \infty).$$

Step 4: State the theorem's conclusion. For any prescribed values $y(3) = \gamma_0$ and $y'(3) = \gamma_1$, there is exactly one solution on:

$$\boxed{(2, \infty)}.$$

The theorem does not guarantee continuation through $x = 2$ because the leading coefficient vanishes there and the normal-form coefficients are singular.

What Happens If The Initial Point Is Singular?

For:

$$(x-2)y'' + (x+1)y' - y = e^x,$$

the point $x = 2$ is singular because the coefficient of y'' is zero.

If initial data are prescribed at $x_0 = 2$, the standard theorem does not apply. This does **not** automatically mean that no solution exists or that solutions are nonunique.

It means only:

the theorem's hypotheses fail, so the theorem makes no promise at that point; the problem requires separate analysis.

Uniqueness And The Zero Solution

Every homogeneous linear equation has the zero solution:

$$y(x) = 0.$$

Its derivatives are also zero, so $L[0] = 0$.

Now consider a regular homogeneous nth-order equation with zero initial data:

$$y(x_0) = y'(x_0) = \cdots = y^{(n-1)}(x_0) = 0.$$

For the zero initial data above, the zero function satisfies both the equation and the data. Uniqueness then implies:

$$\boxed{y(x) \equiv 0}$$

is the only solution of that initial-value problem.

This fact will later help prove that a fundamental set of solutions is independent.

Block 3 Summary

For a regular nth-order linear equation, n initial values at one point select exactly one solution on any interval where the normal-form coefficients and forcing are continuous.

The theorem gives certainty about existence and uniqueness, but it must not be applied across a singular point.

Superposition And The Homogeneous Solution Space

The Goal Of This Block

The goal is to understand why constant linear combinations of homogeneous solutions are again solutions and why an nth-order equation needs n independent solution directions.

superposition does not merely produce more solutions; it reveals the shape of the entire homogeneous solution set.

The Superposition Principle

Suppose y_1 and y_2 solve the same homogeneous linear equation:

$$L[y_1] = 0, \qquad L[y_2] = 0.$$

Let c_1 and c_2 be constants. Operator linearity gives:

$$L[c_1 y_1 + c_2 y_2] = c_1 L[y_1] + c_2 L[y_2].$$

Substitute $L[y_1] = 0$ and $L[y_2] = 0$:

$$\begin{aligned} L[c_1 y_1 + c_2 y_2] &= c_1(0) + c_2(0) \\ &= 0. \end{aligned}$$

Therefore:

$$\boxed{y = c_1 y_1 + c_2 y_2}$$

is also a homogeneous solution.

The same argument works for any finite collection:

$$L\left[\sum_{k=1}^{m} c_k y_k\right] = \sum_{k=1}^{m} c_k L[y_k] = 0.$$

Why The Right Side Must Be Zero

Suppose instead that y_1 and y_2 both solve:

$$L[y] = f(x).$$

For two solutions of $L[y] = f$, the operator values are:

$$L[y_1] = f, \qquad L[y_2] = f.$$

For constants c_1, c_2:

$$\begin{aligned} L[c_1 y_1 + c_2 y_2] &= c_1 L[y_1] + c_2 L[y_2] \\ &= (c_1 + c_2) f. \end{aligned}$$

This equals f only when:

$$c_1 + c_2 = 1$$

or when $f = 0$.

So arbitrary superposition belongs to the homogeneous equation. The nonhomogeneous case has a related but different structure, developed later.

Running Example: Two Oscillatory Solutions

Consider the homogeneous equation:

$$y'' + 4y = 0.$$

Two proposed solutions are:

$$y_1(x) = \cos(2x), \qquad y_2(x) = \sin(2x).$$

Verify y_1 first. Differentiate twice:

$$\begin{aligned} y_1'(x) &= -2\sin(2x), \\ y_1''(x) &= -4\cos(2x). \end{aligned}$$

Substitute y_1 and y_1'' into $y'' + 4y$:

$$\begin{aligned} y_1'' + 4y_1 &= -4\cos(2x) + 4\cos(2x) \\ &= 0. \end{aligned}$$

Now verify y_2. Differentiate twice:

$$\begin{aligned} y_2'(x) &= 2\cos(2x), \\ y_2''(x) &= -4\sin(2x). \end{aligned}$$

Substitute y_2 and y_2'':

$$\begin{aligned} y_2'' + 4y_2 &= -4\sin(2x) + 4\sin(2x) \\ &= 0. \end{aligned}$$

Therefore superposition guarantees that:

$$\boxed{y = c_1\cos(2x) + c_2\sin(2x)}$$

is a solution for every pair of constants c_1, c_2.

We will shortly verify that the two functions are independent. Once that is known, this family is the general solution.

The Solution Set Behaves Like A Vector Space

The solutions of $L[y] = 0$ have two closure properties:

1. adding two solutions gives another solution
2. multiplying a solution by a constant gives another solution

The zero function is also a solution.

These are the defining algebraic behaviours of a vector space. The "vectors" are functions rather than arrows or coordinate lists.

In this viewpoint:

- a solution such as y_1 is one direction
- constants scale that direction
- sums combine directions
- an independent set contains no redundant direction
- a basis is a smallest complete set of directions

This language will make the role of the Wronskian more natural.

Why The Order Predicts The Number Of Directions

A regular nth-order homogeneous linear equation has an n-dimensional solution space.

That means there exist n linearly independent solutions:

$$y_1, y_2, \cdots, y_n,$$

and every homogeneous solution can be written uniquely as:

$$\boxed{y_h = c_1 y_1 + c_2 y_2 + \cdots + c_n y_n}.$$

The n constants correspond to the n initial values:

$$y(x_0), y'(x_0), \ldots, y^{(n-1)}(x_0).$$

The connection is not accidental. Initial data provide n coordinates that identify one point in the n-dimensional solution space.

A Third-Order Illustration

Consider:

$$y''' - y' = 0.$$

Three solutions are:

$$y_1 = 1, \qquad y_2 = e^x, \qquad y_3 = e^{-x}.$$

Verify each solution.

For $y_1 = 1$:

$$y_1' = 0, \qquad y_1''' = 0,$$

For $y_1 = 1$, substitution gives:

$$y_1''' - y_1' = 0 - 0 = 0.$$

For $y_2 = e^x$:

$$y_2' = e^x, \qquad y_2''' = e^x,$$

For $y_2 = e^x$, substitution gives:

$$y_2''' - y_2' = e^x - e^x = 0.$$

For $y_3 = e^{-x}$:

$$y_3' = -e^{-x}, \qquad y_3''' = -e^{-x},$$

For $y_3 = e^{-x}$, substitution gives:

$$y_3''' - y_3' = -e^{-x} - (-e^{-x}) = 0.$$

Once independence is established, the general solution is:

$$\boxed{y_h = c_1 + c_2 e^x + c_3 e^{-x}}.$$

Three genuinely different solution directions are needed because the equation is third-order.

Block 4 Summary

For a homogeneous linear equation:

$$L[y] = 0,$$

constant linear combinations of solutions remain solutions. A regular nth-order equation has n independent solution directions, and their linear combination gives the full homogeneous family.

Linear Independence Without A Determinant

The Goal Of This Block

The goal is to decide whether a collection of functions supplies genuinely different directions or contains redundancy.

We begin with the definition because the Wronskian is a test derived from this idea, not a replacement for understanding it.

The Definition

Functions $y_1, \ldots, y_m$ are **linearly dependent** on an interval I if there are constants $c_1, \ldots, c_m$, not all zero, such that:

$$\boxed{c_1 y_1(x) + \cdots + c_m y_m(x) = 0 \quad \text{for every } x \in I}.$$

They are **linearly independent** on I if the only constants that make this identity true are:

$$c_1 = c_2 = \cdots = c_m = 0.$$

The phrase "for every x in the interval" is essential. An equation that holds at one or two points does not establish dependence.

A Dependent Set With An Explicit Relation

Consider:

$$y_1 = 1 + x, \qquad y_2 = 2 - x, \qquad y_3 = 3.$$

For $y_1 = 1 + x$ and $y_2 = 2 - x$, add the first two functions:

$$\begin{aligned} y_1 + y_2 &= (1 + x) + (2 - x) \\ &= 3 \\ &= y_3. \end{aligned}$$

Move every term to the left:

$$y_1 + y_2 - y_3 = 0.$$

Thus the constants:

$$c_1 = 1, \qquad c_2 = 1, \qquad c_3 = -1$$

are not all zero and satisfy the required identity for every x. Therefore the set is linearly dependent.

The third function adds no new direction; it is already a combination of the first two.

An Independent Polynomial Set

Consider the functions:

$$1, \qquad x, \qquad x^2.$$

Begin with the dependence equation:

$$c_1 + c_2x + c_3x^2 = 0$$

for every x in an interval.

In $c_1 + c_2x + c_3x^2 = 0$, the left side is a polynomial. A polynomial that is zero throughout an interval must have every coefficient equal to zero. Therefore:

$$c_1 = 0, \qquad c_2 = 0, \qquad c_3 = 0.$$

Only the trivial constants work, so:

$$\boxed{\{1, x, x^2\} \text{ is linearly independent.}}$$

A Direct Test For Two Functions

For two functions y_1 and y_2, dependence means one is a constant multiple of the other on the whole interval.

Suppose:

$$c_1y_1 + c_2y_2 = 0$$

with $c_2 \neq 0$. Rearranging gives:

$$y_2 = -\frac{c_1}{c_2}y_1.$$

The ratio is a constant.

For example, e^{3x} and $7e^{3x}$ are dependent because:

$$7e^{3x} - 7e^{3x} = 0$$

The dependence relation for e^{3x} and $7e^{3x}$ can be written as:

$$-7(e^{3x}) + 1(7e^{3x}) = 0.$$

By contrast, e^{3x} and e^{-x} are not constant multiples on any interval of positive length.

Independence Depends On The Interval

Consider:

$$y_1(x) = x, \qquad y_2(x) = |x|.$$

On the interval $(0, \infty)$:

$$|x| = x,$$

so the functions are dependent.

On an interval containing both positive and negative values, suppose:

$$c_1 x + c_2 |x| = 0$$

for every x.

For $x > 0$, $|x| = x$, so:

$$(c_1 + c_2)x = 0,$$

For $x > 0$, the identity $(c_1 + c_2)x = 0$ gives:

$$c_1 + c_2 = 0.$$

For $x < 0$, $|x| = -x$, so:

$$(c_1 - c_2)x = 0,$$

For $x < 0$, the identity $(c_1 - c_2)x = 0$ gives:

$$c_1 - c_2 = 0.$$

Solve the two constant equations:

$$\begin{aligned} c_1 + c_2 &= 0, \\ c_1 - c_2 &= 0. \end{aligned}$$

Adding them gives $2c_1 = 0$, so $c_1 = 0$, and then $c_2 = 0$. Therefore the functions are independent on an interval containing values of both signs.

Why The Number Of Functions Is Not Enough

A second-order equation needs two independent solutions, not merely two listed solutions.

Suppose $y_1 = e^x$ solves a second-order homogeneous equation. Then:

$$y_2 = 5e^x$$

is also a solution by constant scaling.

But:

$$y_2 = 5y_1,$$

Because $y_2 = 5y_1$ for $y_1 = e^x$, the pair is dependent. Its linear combinations reduce to:

$$\begin{aligned} c_1y_1 + c_2y_2 &= c_1e^x + 5c_2e^x \\ &= (c_1 + 5c_2)e^x. \end{aligned}$$

There is still only one effective constant and one direction.

What to remember:

a list reaches the correct length only when none of its functions can be recreated from the others using constant coefficients.

Block 5 Summary

Linear dependence is the existence of a nontrivial constant identity:

$$c_1y_1 + \cdots + c_my_m = 0.$$

Linear independence means that only the all-zero constants work. The identity must hold throughout the stated interval.

The Wronskian As An Independence Test

The Goal Of This Block

The goal is to understand how function values and derivative values can be assembled into a determinant that tests linear independence.

The Wronskian is powerful, but its zero case must be interpreted with care.

The Two-Function Wronskian

For two differentiable functions y_1 and y_2, their Wronskian is:

$$\boxed{W[y_1, y_2](x) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1'}.$$

The columns contain the value and first derivative of each function at the same point.

If the two functions were constant multiples, their columns would be constant multiples as well, forcing the determinant to be zero.

Worked Two-Function Calculation

Find the Wronskian of:

$$y_1 = e^{2x}, \qquad y_2 = e^{-x}.$$

First calculate the derivatives:

$$y_1' = 2e^{2x}, \qquad y_2' = -e^{-x}.$$

For $y_1 = e^{2x}$ and $y_2 = e^{-x}$, insert $y_1' = 2e^{2x}$ and $y_2' = -e^{-x}$ into the determinant:

$$W[y_1, y_2] = \begin{vmatrix} e^{2x} & e^{-x} \\ 2e^{2x} & -e^{-x} \end{vmatrix}.$$

Expand the 2×2 determinant:

$$\begin{aligned} W[y_1, y_2] &= e^{2x}(-e^{-x}) - e^{-x}(2e^{2x}) \\ &= -e^{x} - 2e^{x} \\ &= -3e^{x}. \end{aligned}$$

Since $e^x > 0$ for every real x:

$$W[y_1, y_2](x) \neq 0$$

for every real x. Therefore e^{2x} and e^{-x} are linearly independent on every real interval.

Worked Two-Function Dependent Example

Now consider the pair:

$$y_1 = e^x, \qquad y_2 = 4e^x.$$

Before calculating anything, notice that the second function is a constant multiple of the first:

$$y_2 = 4y_1.$$

We will now see how that repeated direction appears in the Wronskian.

First calculate the derivatives:

$$y_1' = e^x, \qquad y_2' = 4e^x.$$

Insert the functions and their derivatives into the determinant:

$$W[y_1, y_2] = \begin{vmatrix} e^x & 4e^x \\ e^x & 4e^x \end{vmatrix}.$$

Expand the 2×2 determinant:

$$\begin{aligned} W[y_1, y_2] &= e^x(4e^x) - (4e^x)(e^x) \\ &= 4e^{2x} - 4e^{2x} \\ &= 0. \end{aligned}$$

The Wronskian is therefore zero for every real x. The cancellation occurs because the second column of the determinant is four times the first column.

For arbitrary functions, a zero Wronskian on its own is not always enough to prove dependence. In this example, however, we can verify the dependence directly by rearranging $y_2 = 4y_1$:

$$4y_1 - y_2 = 0.$$

The coefficients 4 and -1 are constants and are not both zero. Therefore e^x and $4e^x$ are linearly dependent on every real interval.

The Wronskian For Three Functions

For three twice-differentiable functions:

$$\boxed{W[y_1, y_2, y_3](x) = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix}}.$$

For n functions, continue through derivatives of order $n-1$.

The rows track derivative order. The columns track the individual functions.

Worked Three-Function Calculation

Consider:

$$y_1 = 1, \qquad y_2 = x, \qquad y_3 = e^x.$$

Their first two derivatives are:

	y_1	y_2	y_3
function	1	x	e^x
first derivative	0	1	e^x
second derivative	0	0	e^x

Build the Wronskian:

$$W[1, x, e^x] = \begin{vmatrix} 1 & x & e^x \\ 0 & 1 & e^x \\ 0 & 0 & e^x \end{vmatrix}.$$

For this Wronskian, the upper-triangular diagonal entries are 1, 1, and e^x. Therefore its determinant is their product:

$$\begin{aligned} W[1, x, e^x] &= (1)(1)(e^x) \\ &= e^x. \end{aligned}$$

Since $e^x \neq 0$, the three functions are linearly independent on every real interval.

Why One Nonzero Value Is Enough

Suppose the functions were dependent:

$$c_1 y_1 + \cdots + c_n y_n = 0$$

with constants not all zero.

Differentiate this identity $n - 1$ times. At a point x_0, the constants must satisfy the homogeneous linear system:

$$\begin{pmatrix} y_1(x_0) & \cdots & y_n(x_0) \\ y_1'(x_0) & \cdots & y_n'(x_0) \\ \vdots & & \vdots \\ y_1^{(n-1)}(x_0) & \cdots & y_n^{(n-1)}(x_0) \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

The determinant of the coefficient matrix is the Wronskian $W(x_0)$.

If $W(x_0) \neq 0$, the matrix is invertible, so the only solution is:

$$c_1 = \cdots = c_n = 0.$$

Therefore a nonzero Wronskian at even one point proves independence on the interval.

What A Zero Wronskian Does Not Always Prove

For arbitrary functions, an identically zero Wronskian does not by itself prove dependence.

Consider on $(-1, 1)$:

$$f(x) = x^2, \qquad g(x) = x|x|.$$

For $x > 0$, $g = x^2$, while for $x < 0$, $g = -x^2$. No single constant C makes $g = Cf$ throughout $(-1, 1)$, so the pair is independent there.

Now compute the Wronskian away from $x = 0$.

For $f = x^2$ and $g = x|x|$, when $x > 0$:

$$g = x^2, \qquad g' = 2x,$$

so:

$$W[f, g] = x^2(2x) - x^2(2x) = 0.$$

For $x < 0$:

$$g = -x^2, \qquad g' = -2x,$$

so:

$$\begin{aligned} W[f, g] &= x^2(-2x) - (-x^2)(2x) \\ &= 0. \end{aligned}$$

At $x = 0$, all function and first-derivative values are zero, so $W(0) = 0$ as well.

Thus the Wronskian is identically zero even though the functions are independent on $(-1, 1)$.

The missing hypothesis is important: these functions are not both solutions of one regular second-order homogeneous linear equation on an interval containing 0.

Why Solutions Of One Regular Equation Behave Better

Let y_1 and y_2 solve the same regular second-order homogeneous equation:

$$y'' + p(x)y' + q(x)y = 0,$$

where p and q are continuous on an interval I.

The aim is to show that the Wronskian obeys its own first-order differential equation. Once that equation is known, the value of the Wronskian at one point will control its value throughout the interval.

Because both functions solve this equation, substituting y_1 and y_2 separately gives:

$$y_1'' + p(x)y_1' + q(x)y_1 = 0,$$
$$y_2'' + p(x)y_2' + q(x)y_2 = 0.$$

Solving each equation for its second derivative gives the two substitutions we will need later:

$$y_1'' = -p(x)y_1' - q(x)y_1,$$
$$y_2'' = -p(x)y_2' - q(x)y_2.$$

Now write the Wronskian of the two functions:

$$W = y_1y_2' - y_2y_1'.$$

Differentiate both products using the product rule. For the first product:

$$(y_1y_2')' = y_1'y_2' + y_1y_2'',$$

and for the second product:

$$(y_2 y_1')' = y_2' y_1' + y_2 y_1''.$$

Therefore:

$$\begin{aligned} W' &= (y_1 y_2')' - (y_2 y_1')' \\ &= (y_1' y_2' + y_1 y_2'') - (y_2' y_1' + y_2 y_1'') \\ &= y_1' y_2' + y_1 y_2'' - y_2' y_1' - y_2 y_1''. \end{aligned}$$

The first and third terms cancel because multiplication is commutative:

$$y_1' y_2' = y_2' y_1'.$$

This leaves:

$$W' = y_1 y_2'' - y_2 y_1''.$$

Now substitute the two second-derivative formulas found earlier:

$$W' = y_1 \left[-p(x) y_2' - q(x) y_2\right] - y_2 \left[-p(x) y_1' - q(x) y_1\right].$$

Distribute y_1 and $-y_2$ across the brackets:

$$\begin{aligned} W' = & -p(x) y_1 y_2' - q(x) y_1 y_2 \\ & + p(x) y_2 y_1' + q(x) y_2 y_1. \end{aligned}$$

The two terms containing $q(x)$ cancel because $y_1 y_2 = y_2 y_1$:

$$-q(x)y_1y_2 + q(x)y_2y_1 = 0.$$

Therefore:

$$\begin{aligned} W' &= -p(x)y_1y_2' + p(x)y_2y_1' \\ &= -p(x)\left(y_1y_2' - y_2y_1'\right). \end{aligned}$$

The expression in parentheses is the original Wronskian W. Hence the Wronskian itself satisfies the first-order equation:

$$W' = -p(x)W.$$

To solve it, first move the right side to the left:

$$W' + p(x)W = 0.$$

The next goal is to turn the entire left side into the derivative of one product. Introduce a function $\mu(x)$ and imagine multiplying the equation by it:

$$\mu W' + \mu p(x)W = 0.$$

The product rule says:

$$(\mu W)' = \mu W' + \mu' W.$$

The first term already matches. For the second terms to match as well, choose μ so that:

$$\mu' = p(x)\mu.$$

We now need a function with exactly this derivative property. Define:

$$A(x) = \int_{x_0}^{x} p(s)\, ds.$$

Here s is only a placeholder inside the integral. By the Fundamental Theorem of Calculus:[1]

$$A'(x) = p(x).$$

Now define the **integrating factor** by:

$$\mu(x) = e^{A(x)} = \exp\left(\int_{x_0}^{x} p(s)\, ds\right).$$

Differentiate μ using the chain rule:

$$\begin{aligned} \mu'(x) &= e^{A(x)} A'(x) \\ &= e^{A(x)} p(x) \\ &= p(x)\mu(x). \end{aligned}$$

Thus this choice of μ has exactly the property we required.

[1] If the upper limit moves from x to $x+h$, the added interval has width h and height approximately $p(x)$. Hence $A(x+h) - A(x) \approx p(x)h$, so $(A(x+h) - A(x))/h \approx p(x)$. Taking the limit as $h \to 0$ gives $A'(x) = p(x)$.

Now multiply the original equation $W' + p(x)W = 0$ by $\mu(x)$:

$$\mu\left(W' + p(x)W\right) = 0.$$

Expand the left side:

$$\mu W' + \mu p(x)W = 0.$$

Replace $\mu p(x)$ by μ', since $\mu' = p(x)\mu$:

$$\mu W' + \mu' W = 0.$$

This left side is now the product-rule expansion we wanted:

$$\mu W' + \mu' W = (\mu W)'.$$

Therefore the differential equation becomes:

$$(\mu W)' = 0.$$

A function with zero derivative is constant, so compare its value at x with its value at the chosen point x_0:

$$\mu(x)W(x) = \mu(x_0)W(x_0).$$

At $x = x_0$, the integral in μ has equal limits. It is zero, so:

$$\mu(x_0) = e^0 = 1.$$

Consequently:

$$W(x) = \frac{W(x_0)}{\mu(x)}.$$

Substitute the definition of $\mu(x)$ and move the reciprocal into the exponent. This gives **Abel's identity**:

$$\boxed{W(x) = W(x_0)\exp\left(-\int_{x_0}^{x} p(s)\,ds\right)}.$$

The exponential factor is never zero. Therefore:

- if $W(x_0) \neq 0$ at one point, then $W(x) \neq 0$ everywhere on I
- if $W(x_0) = 0$ at one point, then $W(x) = 0$ everywhere on I

For solutions of the same regular equation, the Wronskian cannot vanish at an isolated point and then recover.

There is one final link to make. Suppose $W(x_0) = 0$. At x_0, the Wronskian matrix is:

$$\begin{pmatrix} y_1(x_0) & y_2(x_0) \\ y_1'(x_0) & y_2'(x_0) \end{pmatrix}.$$

Its determinant is zero, so the matrix is singular. Therefore there are constants c_1 and c_2, not both zero, such that:

$$\begin{aligned} c_1y_1(x_0) + c_2y_2(x_0) &= 0, \\ c_1y_1'(x_0) + c_2y_2'(x_0) &= 0. \end{aligned}$$

Define a new function by combining the two solutions:

$$z = c_1 y_1 + c_2 y_2.$$

Because the original equation is linear and homogeneous, z is also a solution. The two equations above say that its initial data are:

$$z(x_0) = 0, \qquad z'(x_0) = 0.$$

The zero function solves the same equation with the same initial data. Since the equation is regular, the uniqueness theorem says there can be only one such solution. Hence:

$$z(x) = c_1 y_1(x) + c_2 y_2(x) = 0 \qquad \text{for every } x \in I.$$

The constants are not both zero, so this is precisely a linear-dependence relation. This is why an identically zero Wronskian does prove dependence when the functions are solutions of the same regular homogeneous equation.

A Wronskian Decision Rule

For functions $y_1, \dots, y_n$ on an interval:

1. If $W(x_0) \neq 0$ at any point, the functions are independent.
2. If $W \equiv 0$ and the functions are known solutions of the same regular nth-order homogeneous linear equation, they are dependent.
3. If $W \equiv 0$ for arbitrary functions, the test is inconclusive; return to the direct definition.

This is the safe way to use the Wronskian without claiming more than it proves.

Block 6 Summary

The Wronskian packages function values and derivatives into a determinant.

$$W(x_0) \neq 0 \quad \Longrightarrow \quad \text{linear independence.}$$

A zero Wronskian proves dependence only with additional structure, most importantly when the functions solve the same regular homogeneous linear equation.

Fundamental Sets And Initial Conditions

The Goal Of This Block

The goal is to turn a set of known solutions into the general homogeneous solution and then use initial conditions to determine its constants.

a fundamental set is not just a collection of solutions; it is a complete, nonredundant coordinate system for the solution space.

What A Fundamental Set Is

For a regular nth-order homogeneous equation, a **fundamental set of solutions** on an interval I is a collection:

$$\{y_1, y_2, \dots, y_n\}$$

such that:

1. every y_k solves the equation on I
2. the n functions are linearly independent on I

The general homogeneous solution is then:

$$\boxed{y_h = c_1y_1 + c_2y_2 + \cdots + c_ny_n}.$$

Both requirements matter. A set of independent functions is not useful unless the functions solve the equation, and a set of solutions is not fundamental if it contains redundancy.

Completing The Oscillatory Example

Return to:

$$y'' + 4y = 0.$$

We already verified that:

$$y_1 = \cos(2x), \qquad y_2 = \sin(2x)$$

are solutions.

Their derivatives are:

$$y_1' = -2\sin(2x), \qquad y_2' = 2\cos(2x).$$

Build the Wronskian:

$$W[y_1, y_2] = \begin{vmatrix} \cos(2x) & \sin(2x) \\ -2\sin(2x) & 2\cos(2x) \end{vmatrix}.$$

Expand the determinant:

$$\begin{aligned} W[y_1, y_2] &= 2\cos^2(2x) + 2\sin^2(2x) \\ &= 2\left(\cos^2(2x) + \sin^2(2x)\right) \\ &= 2. \end{aligned}$$

Since $W = 2 \neq 0$, the functions are linearly independent. They form a fundamental set, so the general solution is:

$$\boxed{y_h = c_1 \cos(2x) + c_2 \sin(2x)}.$$

Worked Initial-Value Problem

Solve:

$$y'' + 4y = 0, \qquad y(0) = 3, \qquad y'(0) = -4.$$

Step 1: Restate the general solution. The fundamental set gives:

$$y(x) = c_1 \cos(2x) + c_2 \sin(2x).$$

Step 2: Differentiate before applying derivative data. Differentiate the complete family:

$$y'(x) = -2c_1 \sin(2x) + 2c_2 \cos(2x).$$

Step 3: Apply $y(0) = 3$. Substitute $x = 0$ into the complete function formula:

$$3 = c_1 \cos 0 + c_2 \sin 0.$$

Use $\cos 0 = 1$ and $\sin 0 = 0$:

$$3 = c_1.$$

Therefore:

$$c_1 = 3.$$

Step 4: Apply $y'(0) = -4$. Substitute $x = 0$ into the derivative formula:

$$-4 = -2c_1 \sin 0 + 2c_2 \cos 0.$$

Use $\sin 0 = 0$ and $\cos 0 = 1$:

$$-4 = 2c_2.$$

Divide by 2:

$$c_2 = -2.$$

Step 5: Write and verify the selected solution. Insert $c_1 = 3$ and $c_2 = -2$ into the general family:

$$\boxed{y(x) = 3\cos(2x) - 2\sin(2x)}.$$

Differentiate twice:

$$\begin{aligned} y'(x) &= -6\sin(2x) - 4\cos(2x), \\ y''(x) &= -12\cos(2x) + 8\sin(2x). \end{aligned}$$

Substitute into the equation:

$$\begin{aligned} y'' + 4y &= (-12\cos(2x) + 8\sin(2x)) \\ &\quad + 4\,(3\cos(2x) - 2\sin(2x)) \\ &= 0. \end{aligned}$$

The initial values are also visible:

$$y(0) = 3, \qquad y'(0) = -4.$$

The Wronskian Is Also An Initial-Data Determinant

For a second-order family:

$$y = c_1 y_1 + c_2 y_2,$$

the initial conditions at x_0 give:

$$\begin{aligned} c_1 y_1(x_0) + c_2 y_2(x_0) &= \gamma_0, \\ c_1 y_1'(x_0) + c_2 y_2'(x_0) &= \gamma_1. \end{aligned}$$

Write this as a matrix system:

$$\begin{pmatrix} y_1(x_0) & y_2(x_0) \\ y_1'(x_0) & y_2'(x_0) \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} \gamma_0 \\ \gamma_1 \end{pmatrix}.$$

The determinant of the coefficient matrix is:

$$W[y_1, y_2](x_0).$$

If the Wronskian is nonzero, the matrix is invertible, so every pair of initial values determines one and only one pair of constants.

This links three ideas:

$$\boxed{\begin{array}{c}\text{nonzero Wronskian}\\ \Longleftrightarrow\\ \text{independent solution directions}\\ \Longleftrightarrow\\ \text{unique coordinates from initial data}\end{array}}$$

Checking The Third-Order Fundamental Set

For:

$$y''' - y' = 0,$$

we verified the solutions:

$$1, \qquad e^x, \qquad e^{-x}.$$

Their Wronskian is:

$$W[1, e^x, e^{-x}] = \begin{vmatrix} 1 & e^x & e^{-x} \\ 0 & e^x & -e^{-x} \\ 0 & e^x & e^{-x} \end{vmatrix}.$$

Expand along the first column:

$$\begin{aligned} W[1, e^x, e^{-x}] &= 1 \begin{vmatrix} e^x & -e^{-x} \\ e^x & e^{-x} \end{vmatrix} \\ &= e^x e^{-x} - (-e^{-x})e^x \\ &= 1 + 1 \\ &= 2. \end{aligned}$$

The Wronskian is nonzero, so the three solutions form a fundamental set. The general solution is therefore:

$$\boxed{y_h = c_1 + c_2 e^x + c_3 e^{-x}}.$$

Block 7 Summary

A fundamental set for an nth-order homogeneous equation contains exactly n independent solutions. Their Wronskian is nonzero on the regular interval, and their linear combination represents every homogeneous solution.

Initial data determine the coefficients because the same Wronskian matrix is invertible.

The Structure Of Nonhomogeneous Solutions

The Goal Of This Block

The goal is to prove the general form of solutions to:

$$L[y] = f(x)$$

and understand the different roles of the complementary solution and a particular solution.

forcing chooses the location of the solution family; homogeneous solutions describe every direction in which one forced solution may be adjusted.

Associated Homogeneous And Particular Solutions

For the nonhomogeneous equation:

$$L[y] = f(x),$$

the right side $f(x)$ is the forcing. Setting the forcing equal to zero gives the **associated homogeneous equation**:

$$L[y] = 0.$$

Let y_h denote the general solution of this homogeneous equation. Therefore:

$$L[y_h] = 0.$$

Because y_h is the general homogeneous solution, it usually contains the full set of arbitrary constants needed for the order of the equation.

A **particular solution** y_p is one specific function satisfying:

$$L[y_p] = f(x).$$

Unlike y_h, the chosen y_p contains no arbitrary constants. It is one fixed representative of the forced response.

Why $y_h + y_p$ Is A Solution

Suppose:

$$L[y_h] = 0$$

and:

$$L[y_p] = f(x).$$

Apply the linear operator to their sum:

$$\begin{aligned} L[y_h + y_p] &= L[y_h] + L[y_p] \\ &= 0 + f(x) \\ &= f(x). \end{aligned}$$

Since $L[y_h + y_p] = f(x)$, every function of the form:

$$y = y_h + y_p$$

solves the nonhomogeneous equation.

Why This Form Gives Every Solution

It remains to show that no nonhomogeneous solutions are missing.

Let y be any solution of:

$$L[y] = f(x),$$

and let y_p be one chosen particular solution:

$$L[y_p] = f(x).$$

Define their difference:

$$z = y - y_p.$$

Apply L to z:

$$\begin{aligned} L[z] &= L[y - y_p] \\ &= L[y] - L[y_p] \\ &= f(x) - f(x) \\ &= 0. \end{aligned}$$

Because $L[z] = 0$ for $z = y - y_p$, the difference is a homogeneous solution and:

$$y = z + y_p.$$

Rename the arbitrary homogeneous solution z as y_h:

$$\boxed{y = y_h + y_p}.$$

This proves both that the formula works and that it captures every solution.

Complete Nonhomogeneous Example

Consider:

$$y'' - 9y = 18x.$$

We are given that the associated homogeneous equation has independent solutions:

$$e^{3x} \qquad \text{and} \qquad e^{-3x}.$$

Find the general solution and then solve the initial-value problem:

$$y(0) = 1, \qquad y'(0) = 0.$$

Step 1: Build the complementary solution. The associated homogeneous equation is:

$$y'' - 9y = 0.$$

For $y'' - 9y = 0$, the known independent solutions e^{3x} and e^{-3x} give:

$$y_h = c_1 e^{3x} + c_2 e^{-3x}.$$

Step 2: Verify a particular solution. Try the proposed function:

$$y_p = -2x.$$

Its derivatives are:

$$y_p' = -2, \qquad y_p'' = 0.$$

Substitute into the left side $y'' - 9y$:

$$\begin{aligned} y_p'' - 9y_p &= 0 - 9(-2x) \\ &= 18x. \end{aligned}$$

Therefore $y_p = -2x$ is a particular solution.

Step 3: Write the general nonhomogeneous solution. Combine the complementary and particular parts:

$$\boxed{y = c_1 e^{3x} + c_2 e^{-3x} - 2x}.$$

Step 4: Differentiate the complete family. Differentiate before applying the derivative condition:

$$y' = 3c_1 e^{3x} - 3c_2 e^{-3x} - 2.$$

Step 5: Apply $y(0) = 1$. Substitute $x = 0$ into the complete solution:

$$1 = c_1 e^0 + c_2 e^0 - 2(0).$$

Therefore:

$$c_1 + c_2 = 1.$$

Step 6: Apply $y'(0) = 0$. Substitute $x = 0$ into the derivative formula:

$$0 = 3c_1 e^0 - 3c_2 e^0 - 2.$$

Therefore:

$$3c_1 - 3c_2 = 2.$$

Step 7: Solve the constant system visibly. The two equations are:

$$\begin{aligned} c_1 + c_2 &= 1, \\ 3c_1 - 3c_2 &= 2. \end{aligned}$$

Divide the second equation by 3:

$$c_1 - c_2 = \frac{2}{3}.$$

Now add $c_1 + c_2 = 1$ and $c_1 - c_2 = 2/3$:

$$2c_1 = \frac{5}{3}.$$

From $2c_1 = 5/3$, divide by 2:

$$c_1 = \frac{5}{6}.$$

Substitute $c_1 = 5/6$ into $c_1 + c_2 = 1$:

$$\frac{5}{6} + c_2 = 1.$$

Subtract 5/6:

$$c_2 = \frac{1}{6}.$$

The selected solution is:

$$\boxed{y(x) = \frac{5}{6}e^{3x} + \frac{1}{6}e^{-3x} - 2x}.$$

Particular Solutions Are Not Unique

For:

$$y'' - 9y = 18x,$$

we found the particular solution:

$$y_p = -2x.$$

Since e^{3x} solves the homogeneous equation, consider:

$$\tilde{y}_p = -2x + e^{3x}.$$

Apply the operator $L[y] = y'' - 9y$:

$$\begin{aligned} L[\tilde{y}_p] &= L[-2x + e^{3x}] \\ &= L[-2x] + L[e^{3x}] \\ &= 18x + 0 \\ &= 18x. \end{aligned}$$

Thus $\tilde{y}_p$ is also a particular solution.

This does not produce a different general family:

$$\begin{aligned} y_h + \tilde{y}_p &= (c_1 e^{3x} + c_2 e^{-3x}) + (-2x + e^{3x}) \\ &= (c_1 + 1)e^{3x} + c_2 e^{-3x} - 2x. \end{aligned}$$

Since $c_1 + 1$ is still an arbitrary constant, the same family results.

The Geometry Of A Shifted Solution Space

The homogeneous solutions form a vector space because they contain zero and are closed under addition and constant scaling.

The nonhomogeneous solutions usually do not contain zero because:

$$L[0] = 0 \neq f.$$

Instead, the complete nonhomogeneous set is:

$$y_p + \{\text{all homogeneous solutions}\}.$$

It is a shifted copy of the homogeneous solution space. In linear algebra, this is called an **affine space**.

Another way to see it:

> **any two forced solutions differ only by a homogeneous solution, so the homogeneous family measures every allowable adjustment that leaves the forcing unchanged.**

Superposing Different Forcing Terms

Suppose:

$$L[u] = f_1$$

and:

$$L[v] = f_2.$$

Then operator linearity gives:

$$\begin{aligned} L[u+v] &= L[u] + L[v] \\ &= f_1 + f_2. \end{aligned}$$

More generally:

$$L[\alpha u + \beta v] = \alpha f_1 + \beta f_2.$$

This means a complicated forcing can sometimes be split into simpler pieces, with the corresponding particular responses added afterward.

Block 8 Summary

For:

$$L[y] = f,$$

choose one particular solution y_p and find the general homogeneous solution y_h. Then:

$$\boxed{y = y_h + y_p}$$

is the complete nonhomogeneous family. Particular solutions are not unique, but any two differ by a homogeneous solution.

A Reliable Theory Workflow And Common Mistakes

The Goal Of This Block

The goal is to decide which theorem or structural result applies without skipping its hypotheses.

This chapter is less about mechanical solving and more about making logically complete conclusions from the information given.

A Theory-First Workflow

Step 1: Classify the equation. State its order, linearity, homogeneity, and coefficient type.

Step 2: Identify a regular interval. Check the leading coefficient and continuity of the normal-form coefficients.

Step 3: Define the operator. Write the equation as $L[y] = f$ and its associated homogeneous equation as $L[y] = 0$.

Step 4: Verify every proposed solution. Differentiate and substitute into the relevant equation.

Step 5: Test independence. Use the direct definition or the Wronskian with the correct hypotheses.

Step 6: Build the general family. Use a fundamental homogeneous set, then add one particular solution if the equation is forced.

Step 7: Apply initial data. Differentiate the complete family as needed, display the resulting constant system, and solve it visibly.

Step 8: State the interval and conclusion. Say exactly what is known: a solution, a general solution, independence, or a uniqueness guarantee.

Common Mistake: Confusing Homogeneous With Nonlinear

The equation:

$$y'' + xy' - 4y = 0$$

is linear and homogeneous.

The equation:

$$y'' + y^2 = 0$$

has a zero right side but is nonlinear. In this chapter, the homogeneous solution-space theory applies only to linear equations.

Common Mistake: Treating Any Two Solutions As A Basis

For a second-order equation, two solutions form a fundamental set only if they are independent.

The pair:

$$e^x, \qquad 4e^x$$

contains one direction twice. Its Wronskian is zero, and no choice of two constants can create a second independent shape.

Common Mistake: Using Variable Multipliers In Superposition

If y_1 and y_2 solve a homogeneous linear equation, then:

$$c_1 y_1 + c_2 y_2$$

is a solution for constants c_1, c_2.

The expression:

$$u(x)y_1 + v(x)y_2$$

does not follow from ordinary superposition because differentiating it creates product-rule terms involving u' and v'.

Variable coefficients are used deliberately in methods such as variation of parameters, but they must satisfy additional equations.

Common Mistake: Reading Too Much From $W = 0$

For arbitrary functions:

$$W \equiv 0$$

does not always prove dependence.

Before drawing that conclusion, ask whether the functions are known solutions of the same regular homogeneous linear equation on the interval. If not, use the direct constant-combination definition.

Common Mistake: Adding Two Particular Solutions

If $L[y_{p1}] = f$ and $L[y_{p2}] = f$, then:

$$L[y_{p1} + y_{p2}] = 2f,$$

not f.

To generate another solution for the same forcing, add a homogeneous solution:

$$L[y_p + y_h] = f + 0 = f.$$

Common Mistake: Omitting The Interval

Existence, uniqueness, and independence are interval statements.

If a leading coefficient vanishes or a normal-form coefficient is discontinuous, the theorem may apply on separate intervals but not across the singular point.

Write the interval beside the final conclusion.

Chapter Method Summary

The chapter's structural chain is:

$$\begin{aligned}
&\text{linear operator}\\
&\implies \text{superposition for } L[y]=0\\
&\implies \text{homogeneous solution space}\\
&\implies \text{independent fundamental set}\\
&\implies y_h = c_1y_1 + \cdots + c_ny_n\\
&\implies y = y_h + y_p \text{ for } L[y]=f.
\end{aligned}$$

The central lesson is:

> **verify the operator, interval, and independence before calling a linear combination the general solution.**

Original Practice Set

Practice Problems

Problem 1: Classify Linear Equations. For each equation, state its order and whether it is linear. For each linear equation, also state whether it is homogeneous and whether its coefficients are constant or variable.

$$\text{(a)} \qquad (1+t^2)u''' - tu' + u = 0,$$

$$\text{(b)} \qquad u'' + uu' = \cos t,$$

$$\text{(c)} \qquad (t-1)u'' + e^t u = 3.$$

For part (c), identify the maximal regular intervals.

Problem 2: Use Operator Linearity. Suppose L is linear and:

$$L[u] = x, \qquad L[v] = e^x.$$

Find:

$$L[2u - 3v].$$

Problem 3: Apply Existence And Uniqueness. Consider:

$$(x+4)y'' + (\sin x)y' + y = x^2.$$

For initial conditions prescribed at $x_0 = 0$, state the largest interval on which the standard linear existence-and-uniqueness theorem guarantees one solution. Explain what the theorem says if the initial point is $x_0 = -4$.

Problem 4: Verify Superposition. Verify that:

$$y_1 = \cos(5x), \qquad y_2 = \sin(5x)$$

solve:

$$y'' + 25y = 0.$$

Then explain why $4y_1 - 3y_2$ is also a solution.

Problem 5: Prove Dependence Directly. Show that the set:

$$\{2 + x,\ 1 - 2x,\ 5\}$$

is linearly dependent by displaying constants, not all zero, that produce the zero function.

Problem 6: Prove Independence Directly. Use the direct definition and differentiation to show that:

$$\{1 + x,\ e^x\}$$

is linearly independent on every real interval of positive length.

Problem 7: Calculate A Two-Function Wronskian. Find the Wronskian of:

$$y_1 = e^x, \qquad y_2 = xe^x.$$

State the independence conclusion.

Problem 8: Interpret A Wronskian With A Zero. Calculate the Wronskian of:

$$1, \qquad x, \qquad x^3.$$

The Wronskian vanishes at $x = 0$. Decide whether the functions are independent on $(-\infty, \infty)$ and explain why the zero at one point does not overturn your conclusion.

Problem 9: Build A Third-Order General Solution. For:

$$y''' - 4y' = 0,$$

the functions:

$$1, \qquad e^{2x}, \qquad e^{-2x}$$

are known solutions. Calculate their Wronskian and use it to write the general solution.

Problem 10: Apply Initial Conditions To A Fundamental Set. Solve:

$$y'' + 25y = 0, \qquad y(0) = 2, \qquad y'(0) = 15,$$

using the fundamental set $\{\cos(5x), \sin(5x)\}$.

Problem 11: Add A Particular Solution. For:

$$y'' - 16y = 8x,$$

the associated homogeneous equation has fundamental set:

$$\{e^{4x}, e^{-4x}\}.$$

Verify that $y_p = -x/2$ is a particular solution and write the complete general solution.

Problem 12: Compare Two Particular Solutions. Let:

$$L[y] = y'' + y.$$

Show that:

$$y_{p1} = x^2 \qquad \text{and} \qquad y_{p2} = x^2 + \sin x$$

solve the same nonhomogeneous equation. Identify that equation and explain why the two particular solutions differ by a homogeneous solution.

Practice Answers

Answer 1. For part (a):

$$(1 + t^2)u''' - tu' + u = 0.$$

The highest derivative is u''', so the equation is third-order. The unknown and its derivatives appear only to the first power, with coefficients depending only on t, so it is linear.

The right side is zero, so it is homogeneous. The coefficients $1 + t^2$ and $-t$ vary with t, so it has variable coefficients.

The complete classification for part (a) is:

$$\boxed{\text{third-order, linear, homogeneous, variable-coefficient.}}$$

For part (b):

$$u'' + uu' = \cos t.$$

The highest derivative is u'', so it is second-order. The product uu' makes it nonlinear.

Thus part (b) is:

$$\boxed{\text{second-order and nonlinear.}}$$

For part (c):

$$(t-1)u'' + e^t u = 3.$$

The highest derivative is u'', so it is second-order. The unknown and its derivatives appear linearly. Since the forcing is $3 \neq 0$, it is nonhomogeneous. Its coefficients vary with t.

Thus part (c) is:

$$\boxed{\text{second-order, linear, nonhomogeneous, variable-coefficient.}}$$

The leading coefficient $t-1$ vanishes at $t = 1$. Therefore the maximal regular intervals are:

$$\boxed{(-\infty, 1) \quad \text{and} \quad (1, \infty)}.$$

Answer 2. Operator linearity gives:

$$L[2u - 3v] = 2L[u] - 3L[v].$$

Substitute $L[u] = x$ and $L[v] = e^x$:

$$\begin{aligned} L[2u - 3v] &= 2(x) - 3(e^x) \\ &= 2x - 3e^x. \end{aligned}$$

Therefore:

$$\boxed{L[2u - 3v] = 2x - 3e^x}.$$

Answer 3. Start from:

$$(x + 4)y'' + (\sin x)y' + y = x^2.$$

For $x \neq -4$, divide every term by $x + 4$:

$$y'' + \frac{\sin x}{x + 4}y' + \frac{1}{x + 4}y = \frac{x^2}{x + 4}.$$

The normal-form coefficients and forcing are continuous on:

$$(-\infty, -4) \qquad \text{and} \qquad (-4, \infty).$$

Since $x_0 = 0$ lies in $(-4, \infty)$, the theorem guarantees exactly one solution on:

$$\boxed{(-4, \infty)}.$$

If $x_0 = -4$, the leading coefficient is zero and normal form is singular. The standard theorem makes no guarantee there. This failure of the hypotheses does not by itself prove nonexistence or nonuniqueness.

Answer 4. For:

$$y_1 = \cos(5x),$$

differentiate twice:

$$y_1' = -5\sin(5x), \qquad y_1'' = -25\cos(5x).$$

Substitute into $y'' + 25y$:

$$y_1'' + 25y_1 = -25\cos(5x) + 25\cos(5x) = 0.$$

For:

$$y_2 = \sin(5x),$$

differentiate twice:

$$y_2' = 5\cos(5x), \qquad y_2'' = -25\sin(5x).$$

Substitute:

$$y_2'' + 25y_2 = -25\sin(5x) + 25\sin(5x) = 0.$$

Thus $L[y_1] = 0$ and $L[y_2] = 0$ for $L[y] = y'' + 25y$. By linearity:

$$\begin{aligned} L[4y_1 - 3y_2] &= 4L[y_1] - 3L[y_2] \\ &= 4(0) - 3(0) \\ &= 0. \end{aligned}$$

Therefore $4\cos(5x) - 3\sin(5x)$ is also a solution.

Answer 5. Let:

$$f_1 = 2 + x, \qquad f_2 = 1 - 2x, \qquad f_3 = 5.$$

Calculate $2f_1 + f_2$:

$$\begin{aligned} 2f_1 + f_2 &= 2(2 + x) + (1 - 2x) \\ &= 4 + 2x + 1 - 2x \\ &= 5 \\ &= f_3. \end{aligned}$$

Therefore:

$$2f_1 + f_2 - f_3 = 0.$$

The constants $2, 1, -1$ are not all zero, so the set is linearly dependent.

Answer 6. Assume constants c_1, c_2 satisfy:

$$c_1(1 + x) + c_2 e^x = 0$$

for every x in a real interval of positive length.

Differentiate once:

$$c_1 + c_2 e^x = 0.$$

Differentiate a second time:

$$c_2 e^x = 0.$$

The second derivative identity is $c_2 e^x = 0$. Since $e^x \neq 0$:

$$c_2 = 0.$$

Substitute $c_2 = 0$ into $c_1 + c_2 e^x = 0$:

$$c_1 = 0.$$

Only the trivial constants work. Therefore:

$$\boxed{\{1 + x, e^x\} \text{ is linearly independent.}}$$

Answer 7. For:

$$y_1 = e^x, \qquad y_2 = xe^x,$$

For $y_1 = e^x$ and $y_2 = xe^x$, the derivatives are:

$$y_1' = e^x, \qquad y_2' = e^x + xe^x.$$

Build and expand the Wronskian:

$$\begin{aligned} W[y_1, y_2] &= \begin{vmatrix} e^x & xe^x \\ e^x & e^x + xe^x \end{vmatrix} \\ &= e^x(e^x + xe^x) - xe^x(e^x) \\ &= e^{2x} + xe^{2x} - xe^{2x} \\ &= e^{2x}. \end{aligned}$$

Since $e^{2x} > 0$ for every real x, the functions are linearly independent on every real interval.

Answer 8. For the functions $1, x, x^3$, construct the derivative table:

	1	x	x^3
function	1	x	x^3
first derivative	0	1	$3x^2$
second derivative	0	0	$6x$

The Wronskian is:

$$W[1, x, x^3] = \begin{vmatrix} 1 & x & x^3 \\ 0 & 1 & 3x^2 \\ 0 & 0 & 6x \end{vmatrix}.$$

This matrix is upper triangular, so:

$$\boxed{W[1, x, x^3] = 6x}.$$

Although $W(0) = 0$, we have, for example:

$$W(1) = 6 \neq 0.$$

A nonzero Wronskian at one point proves independence on the interval. Therefore $1, x, x^3$ are linearly independent on $(-\infty, \infty)$.

Answer 9. For:

$$y_1 = 1, \qquad y_2 = e^{2x}, \qquad y_3 = e^{-2x},$$

the first two derivatives are:

	y_1	y_2	y_3
function	1	e^{2x}	e^{-2x}
first derivative	0	$2e^{2x}$	$-2e^{-2x}$
second derivative	0	$4e^{2x}$	$4e^{-2x}$

For $1, e^{2x}, e^{-2x}$, build the Wronskian from the derivative table:

$$W = \begin{vmatrix} 1 & e^{2x} & e^{-2x} \\ 0 & 2e^{2x} & -2e^{-2x} \\ 0 & 4e^{2x} & 4e^{-2x} \end{vmatrix}.$$

Expand along the first column:

$$\begin{aligned} W &= \begin{vmatrix} 2e^{2x} & -2e^{-2x} \\ 4e^{2x} & 4e^{-2x} \end{vmatrix} \\ &= (2e^{2x})(4e^{-2x}) - (-2e^{-2x})(4e^{2x}) \\ &= 8 + 8 \\ &= 16. \end{aligned}$$

Since $W = 16 \neq 0$, the three known solutions form a fundamental set for the third-order equation. Therefore:

$$\boxed{y_h = c_1 + c_2 e^{2x} + c_3 e^{-2x}}.$$

Answer 10. The fundamental set gives the general solution:

$$y = c_1 \cos(5x) + c_2 \sin(5x).$$

Differentiate the complete family:

$$y' = -5c_1 \sin(5x) + 5c_2 \cos(5x).$$

Use $y(0) = 2$ in $y = c_1 \cos(5x) + c_2 \sin(5x)$:

$$2 = c_1 \cos 0 + c_2 \sin 0 = c_1.$$

Thus:

$$c_1 = 2.$$

Use $y'(0) = 15$ in the derivative formula:

$$15 = -5c_1 \sin 0 + 5c_2 \cos 0 = 5c_2.$$

Divide by 5:

$$c_2 = 3.$$

Therefore the selected solution is:

$$\boxed{y = 2\cos(5x) + 3\sin(5x)}.$$

Answer 11. The associated homogeneous fundamental set gives:

$$y_h = c_1 e^{4x} + c_2 e^{-4x}.$$

For the proposed particular solution:

$$y_p = -\frac{1}{2}x,$$

For $y_p = -x/2$, the derivatives are:

$$y_p' = -\frac{1}{2}, \qquad y_p'' = 0.$$

Substitute into $y'' - 16y$:

$$\begin{aligned} y_p'' - 16y_p &= 0 - 16\left(-\frac{1}{2}x\right) \\ &= 8x. \end{aligned}$$

Thus y_p is a particular solution. Add it to the complete homogeneous family:

$$\boxed{y = c_1 e^{4x} + c_2 e^{-4x} - \frac{1}{2}x}.$$

Answer 12. The operator is:

$$L[y] = y'' + y.$$

For $y_{p1} = x^2$:

$$y_{p1}'' = 2.$$

Therefore:

$$L[y_{p1}] = 2 + x^2.$$

For $y_{p2} = x^2 + \sin x$:

$$y_{p2}'' = 2 - \sin x.$$

Therefore:

$$\begin{aligned} L[y_{p2}] &= (2 - \sin x) + (x^2 + \sin x) \\ &= 2 + x^2. \end{aligned}$$

Both functions solve:

$$\boxed{y'' + y = x^2 + 2}.$$

Their difference is:

$$y_{p2} - y_{p1} = \sin x.$$

Since:

$$(\sin x)'' + \sin x = -\sin x + \sin x = 0,$$

the difference is a solution of the associated homogeneous equation.

Final Takeaway

Linear solution theory is organised by a small number of connected ideas:

$$\begin{aligned} L[\alpha u + \beta v] &= \alpha L[u] + \beta L[v], \\ L[y_h] &= 0, \\ W[y_1, \dots, y_n] \neq 0 &\implies \{y_1, \dots, y_n\} \text{ is independent}, \\ y_h &= c_1 y_1 + \cdots + c_n y_n, \\ L[y] = f &\implies y = y_h + y_p. \end{aligned}$$

The chapter's final lesson is:

the general solution is justified by three checks: every basis function solves the homogeneous equation, the basis functions are independent, and any particular term reproduces the forcing exactly.